US012401243B2

(12) United States Patent
Rapant et al.

(10) Patent No.: US 12,401,243 B2
(45) Date of Patent: Aug. 26, 2025

(54) ROTOR ASSEMBLY FOR AN ELECTRIC MOTOR

(71) Applicant: **MILWAUKEE ELECTRIC TOOL CORPORATION**, Brookfield, WI (US)

(72) Inventors: **Russell M. Rapant**, Menomonee Falls, WI (US); **Andrew T. Beyerl**, Pewaukee, WI (US)

(73) Assignee: **MILWAUKEE ELECTRIC TOOL CORPORATION**, Brookfield, WI (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/364,903**

(22) Filed: **Aug. 3, 2023**

(65) **Prior Publication Data**

US 2023/0378834 A1 Nov. 23, 2023

Related U.S. Application Data

(62) Division of application No. 16/698,283, filed on Nov. 27, 2019, now Pat. No. 11,742,710.

(60) Provisional application No. 62/772,924, filed on Nov. 29, 2018.

(51) **Int. Cl.**

H02K 1/28	(2006.01)
H02K 1/17	(2006.01)
H02K 15/12	(2006.01)

(52) **U.S. Cl.**

CPC ***H02K 1/28*** (2013.01); ***H02K 1/17*** (2013.01); ***H02K 15/12*** (2013.01); *Y10T 29/49012* (2015.01); *Y10T 29/53143* (2015.01)

(58) **Field of Classification Search**

CPC .. H02K 9/06; H02K 5/18; H02K 9/04; H02K 1/27; H02K 1/28; H02K 2201/06; Y10T 29/49012; Y10T 29/49009; Y10T 29/49071; Y10T 29/49073; Y10T 29/53143

USPC 29/598, 557, 558, 592, 596, 597, 605, 29/606, 729, 732, 738

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,883,408 A	11/1989	Borcherding
5,329,199 A	7/1994	Yockey et al.
5,536,985 A	7/1996	Ward et al.

(Continued)

FOREIGN PATENT DOCUMENTS

CN	201307810 Y	9/2009
CN	204243940 U	4/2015

(Continued)

OTHER PUBLICATIONS

European Patent Office Action for Application No. 19888772.1 dated May 31, 2024 (6 pages).

(Continued)

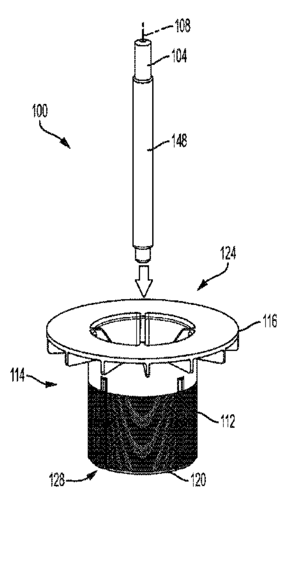
Primary Examiner — Thiem D Phan

(74) *Attorney, Agent, or Firm* — Michael Best & Friedrich LLP

(57) **ABSTRACT**

A method of forming a molded rotor assembly for an electric motor includes providing a lamination stack, molding an insulative material to the lamination stack to form an integral fan and magnet retention coupled to the lamination stack, the lamination stack and the integral fan and magnet retention together forming a rotor body, and pressing a shaft into a central aperture formed in the rotor body to achieve an interference fit between the shaft and the lamination stack.

20 Claims, 7 Drawing Sheets



(56) **References Cited**

U.S. PATENT DOCUMENTS

6,735,846 B2	5/2004	Du	
6,819,022 B2	11/2004	Yamamoto et al.	
6,946,758 B2	9/2005	Du et al.	
7,013,552 B2	3/2006	Du	
7,096,566 B2	8/2006	Du et al.	
7,215,048 B2	5/2007	Du et al.	
7,464,455 B2	12/2008	Du et al.	
7,591,063 B2	9/2009	West	
7,685,697 B2	3/2010	Du et al.	
7,814,641 B2	10/2010	Du et al.	
7,847,457 B2	12/2010	Achor	
8,203,239 B2	6/2012	Du et al.	
8,291,574 B2	10/2012	Achor	
8,324,764 B2	12/2012	Du et al.	
8,850,690 B2	10/2014	Du et al.	
8,866,353 B2 *	10/2014	Miyama	H02K 11/33 310/58
8,896,176 B2	11/2014	Ryu et al.	
8,901,787 B2	12/2014	Du et al.	
8,937,412 B2	1/2015	Du et al.	
8,987,964 B2	3/2015	Achor	
8,997,332 B2	4/2015	Du et al.	
9,472,989 B2	10/2016	Du et al.	
9,755,490 B2	9/2017	Inuzuka	
10,205,365 B2	2/2019	Beyerl et al.	
10,276,067 B2 *	4/2019	Nadig	G09B 23/188
2006/0055264 A1	3/2006	Johnson et al.	
2006/0261694 A1	11/2006	Chien-Hsiung et al.	
2006/0273679 A1	12/2006	Iwase et al.	
2007/0236091 A1	10/2007	Fukushima	
2011/0293448 A1	12/2011	Ishiguro	
2012/0183417 A1	7/2012	Linnenbrock et al.	
2014/0042834 A1	2/2014	Asahi et al.	
2014/0125158 A1	5/2014	Hessenberger et al.	
2016/0181891 A1	6/2016	Fogle et al.	
2016/0197535 A1	7/2016	De Filippis et al.	
2017/0207669 A1	7/2017	Zhang et al.	
2017/0288499 A1	10/2017	Beyerl et al.	
2017/0346364 A1	11/2017	Hessenberger et al.	
2017/0373569 A1	12/2017	Fung et al.	
2020/0177041 A1	6/2020	Rapant et al.	

FOREIGN PATENT DOCUMENTS

CN	205489850	U	8/2016
CN	107147256	A	9/2017
CN	207304189	U	5/2018
CN	108494127	A	9/2018
DE	102007060011	A1	7/2009
EP	2824810	A1	1/2015
EP	2963775	A1	1/2016
JP	S631341	A	1/1988
JP	H04190656	A	7/1992
JP	H0914174	A	1/1997
JP	2007209178	A	8/2007
JP	2012139070	A	7/2012
KR	2019990036381	U	9/1999
KR	1020030023279	A	3/2003
KR	1020180020041	A	2/2018
KR	101842760	B1	3/2018
WO	2018103969	A1	6/2018

OTHER PUBLICATIONS

Partial Supplementary European Search Report for Application No. 21841695.6 dated Jul. 11, 2024 (18 pages).

International Search Report and Written Opinion for Application No. PCT/US2019/063682 dated Mar. 23, 2020 (7 pages).

International Search Report and Written Opinion for Application No. PCT/US2021/041363 dated Oct. 29, 2021 (14 pages).

Extended European Search Report for Application No. 19888772.1 dated Jul. 29, 2022 (10 pages).

Chinese Patent Office Action for Application No. 201980090493.3 dated Sep. 8, 2024 (17 pages including machine English translation).

Extended European Search Report for Application No. 21841695.6 dated Oct. 2, 2024 (17 pages).

Zhou Weiping, “Mechanical Manufacturing Technology,” Huazhong University of Science and Technology Press, Aug. 2002, p. 300 (5 total pages including English translation).

Chinese Patent Office Action for Application No. 201980090493.3 dated Mar. 6, 2025 (18 pages including machine English translation).

* cited by examiner

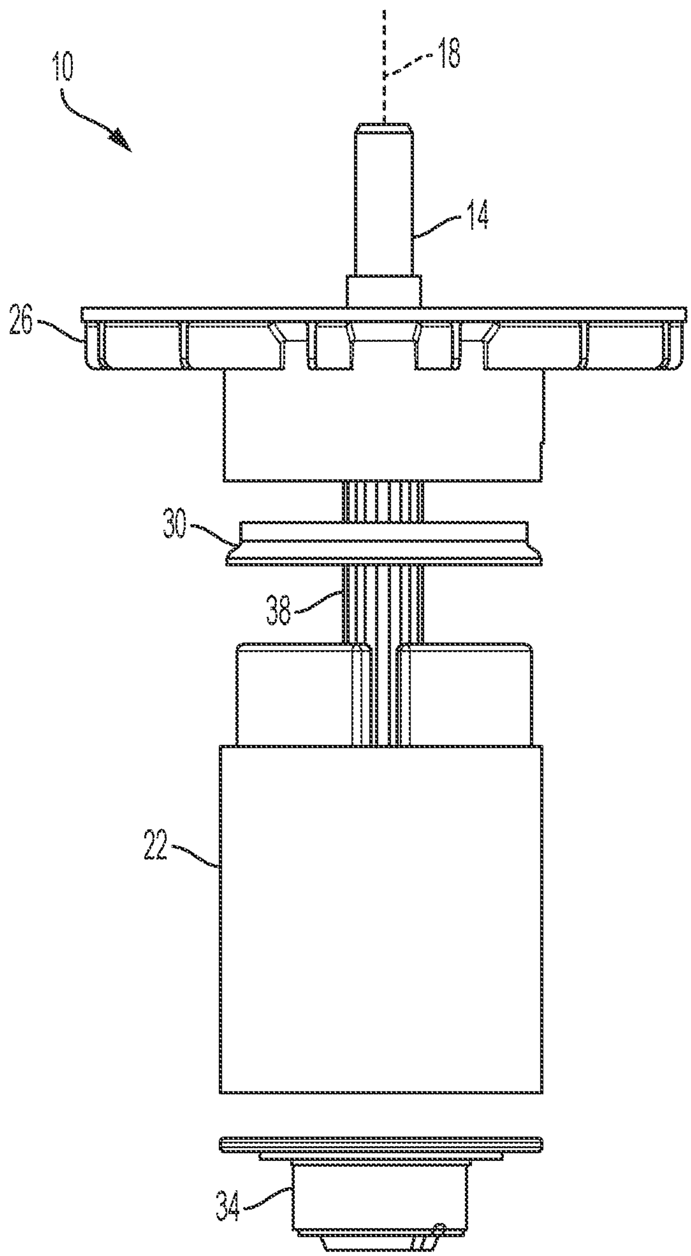


FIG. 1
PRIOR ART

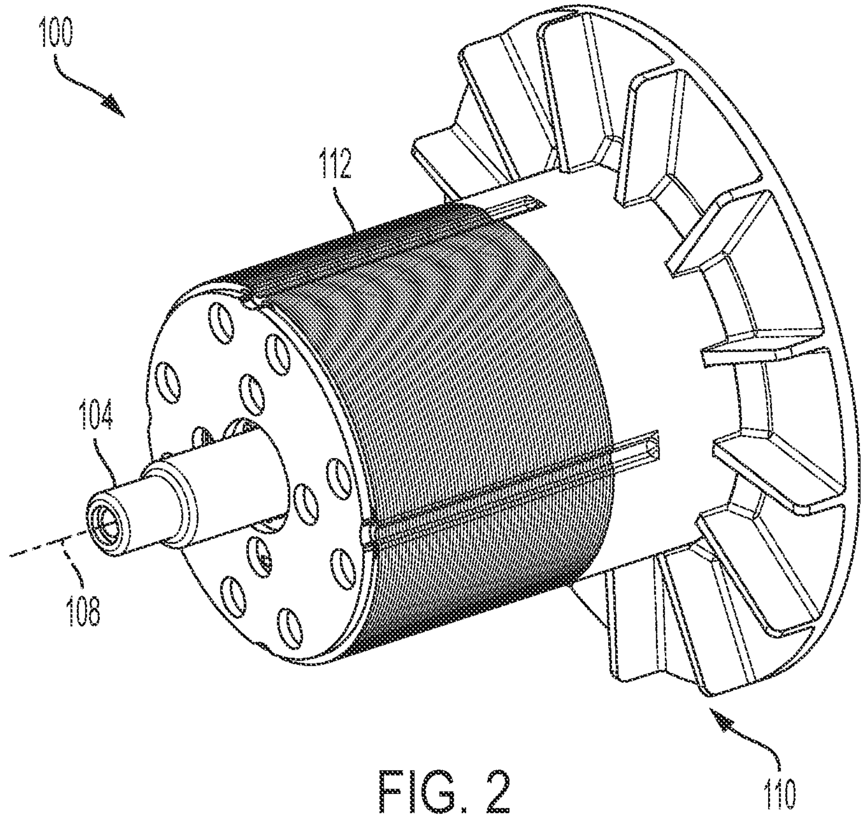


FIG. 2

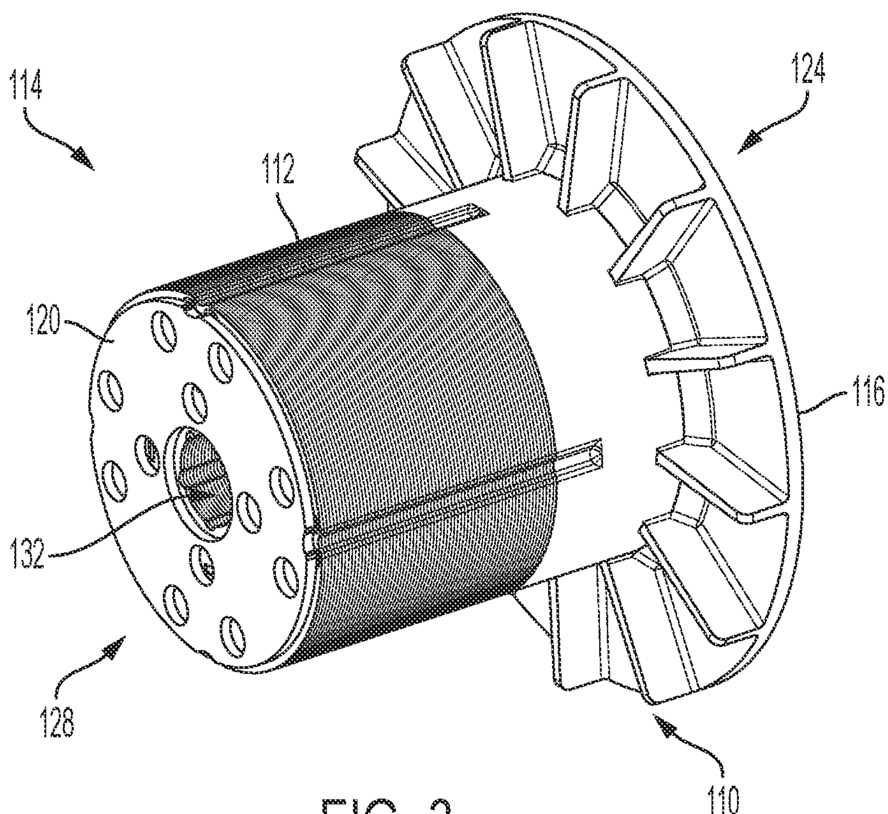


FIG. 3

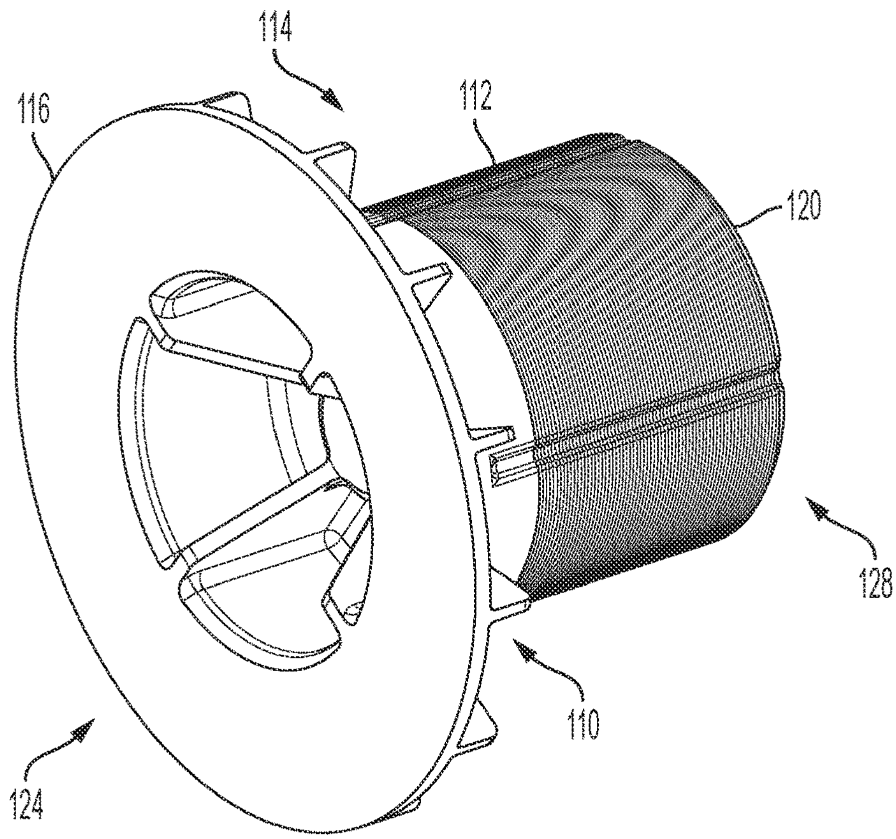


FIG. 4

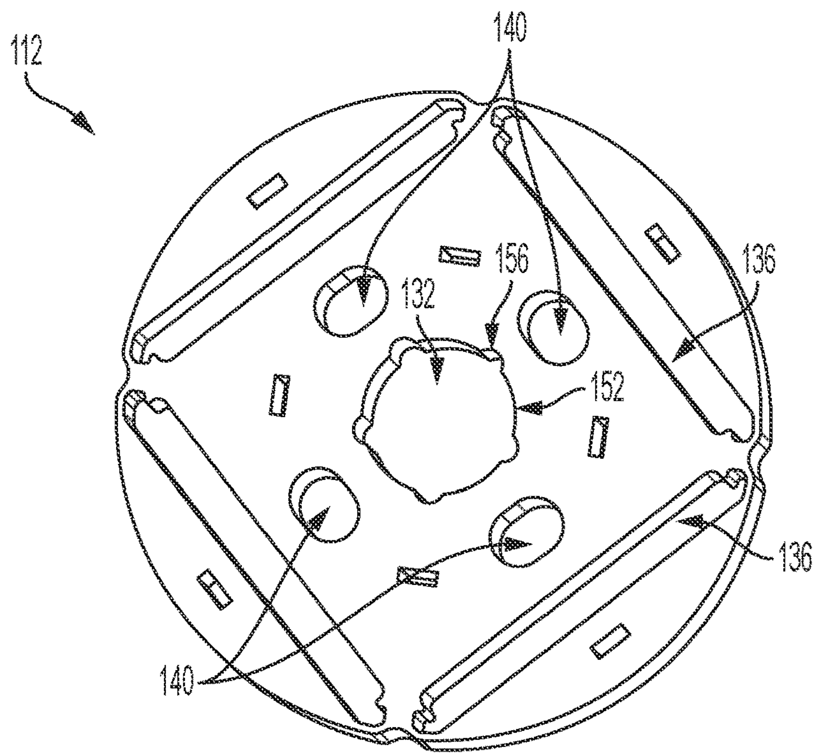


FIG. 5

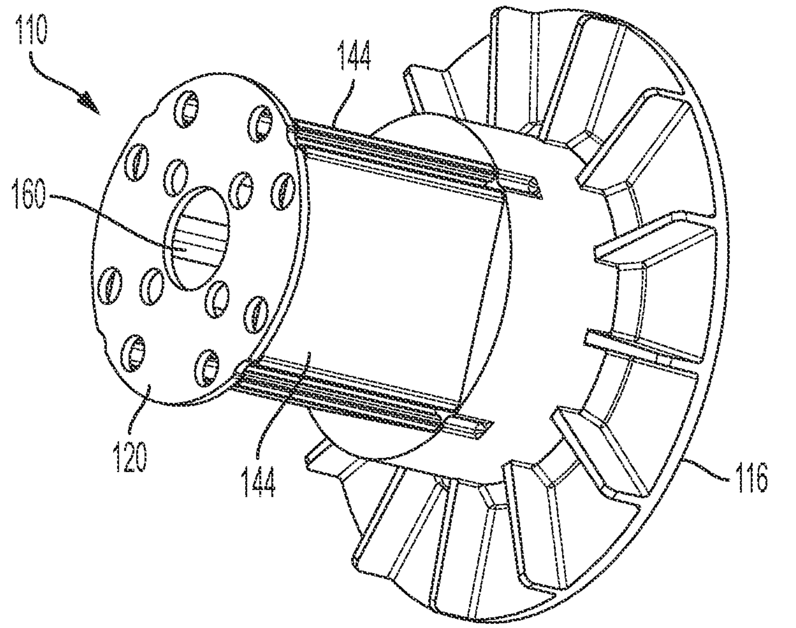


FIG. 6

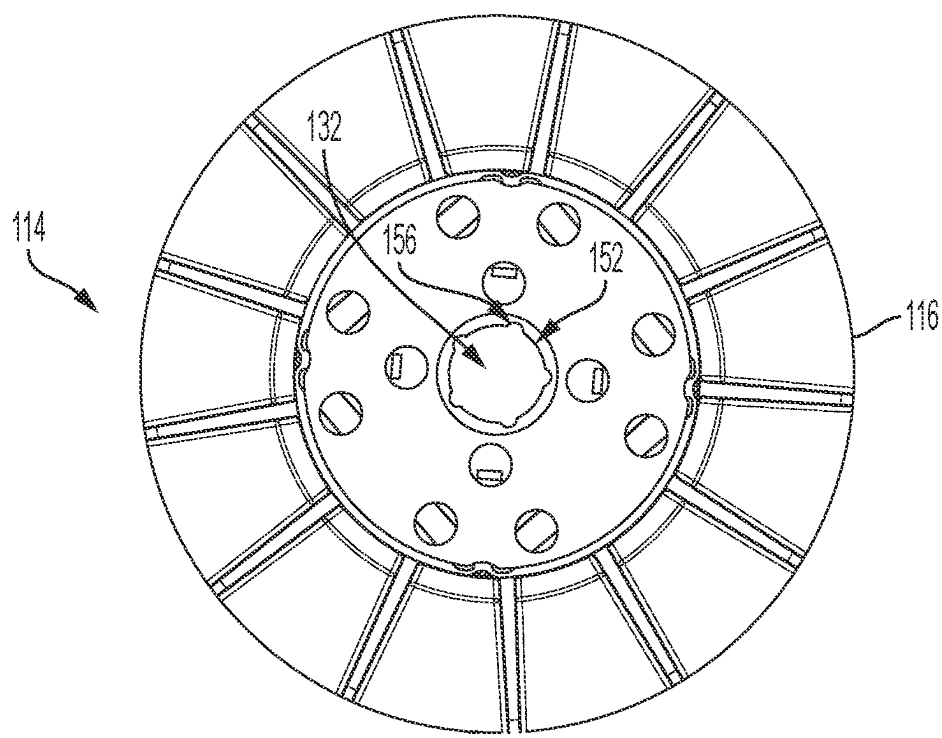


FIG. 7

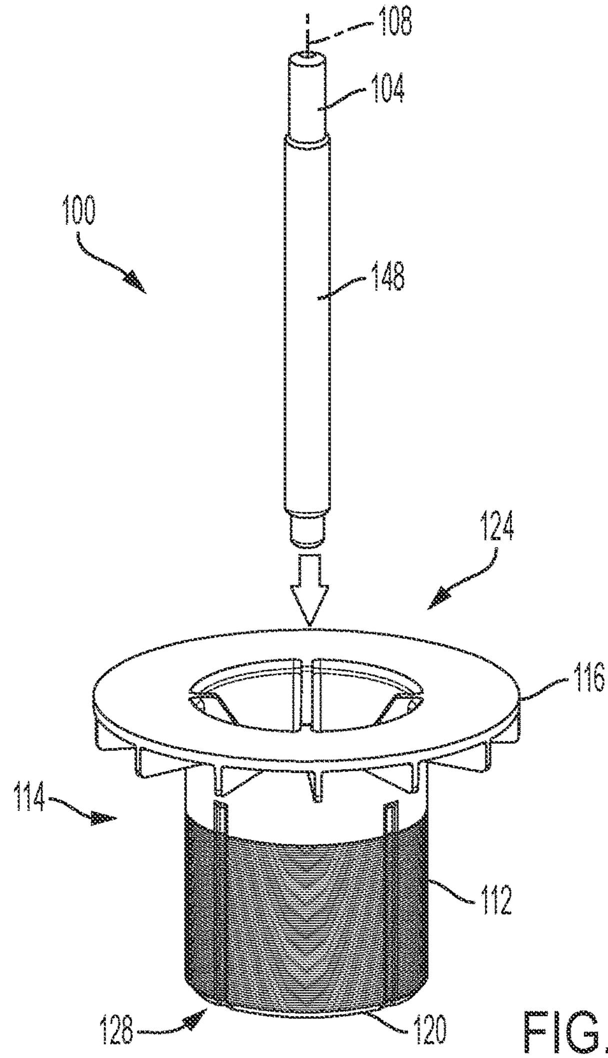

108
104
100
148
124
116
114
112
128
120

FIG. 8

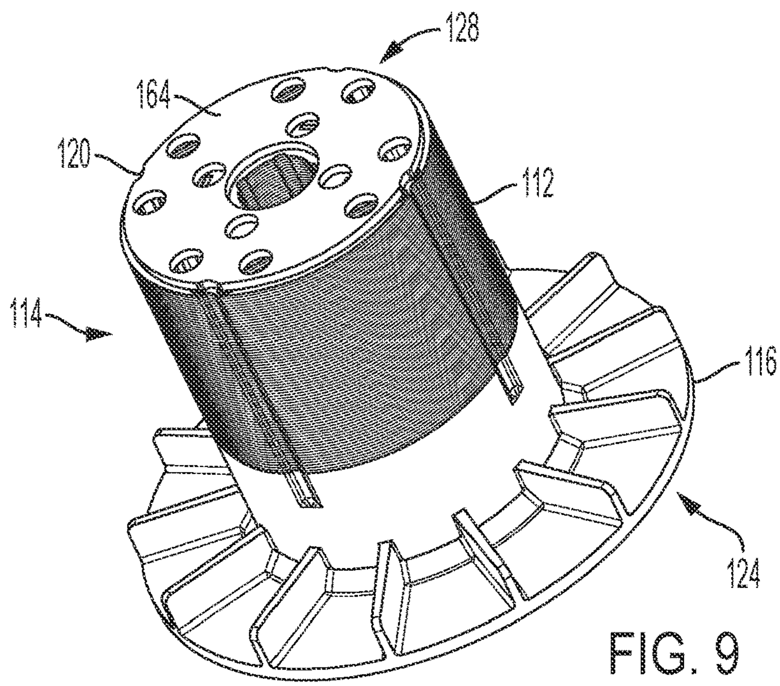

128
164
120
112
114
116
124

FIG. 9

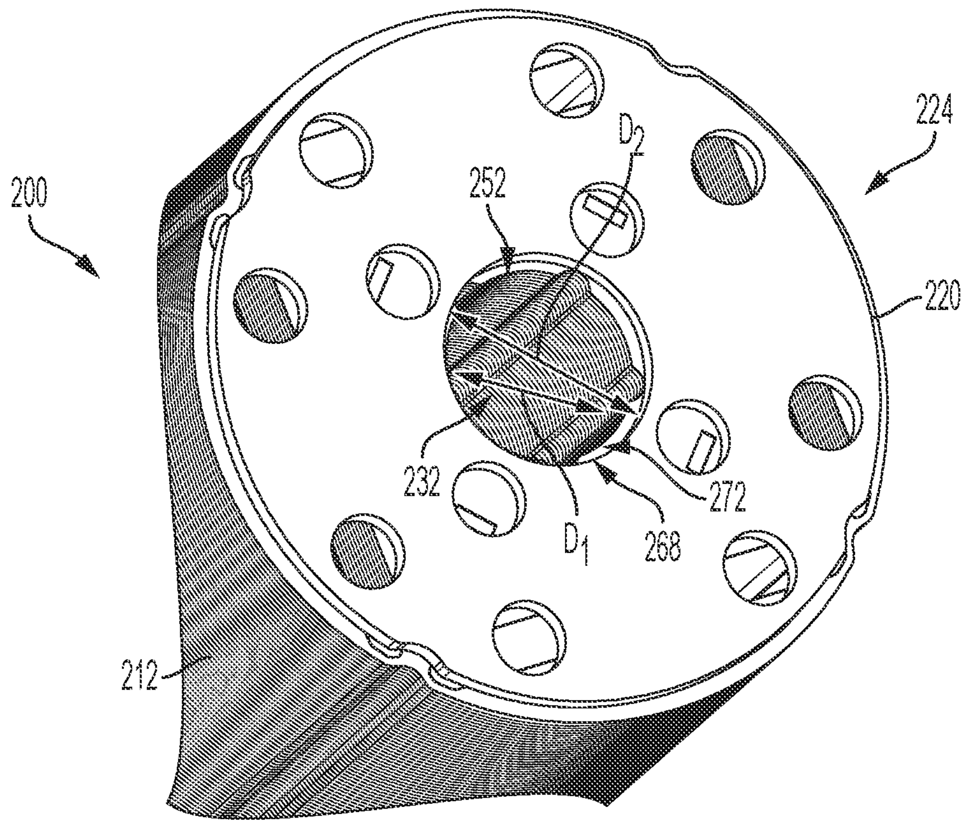

224
200
D_2
252
220
232
272
268
D_1
212

FIG. 10

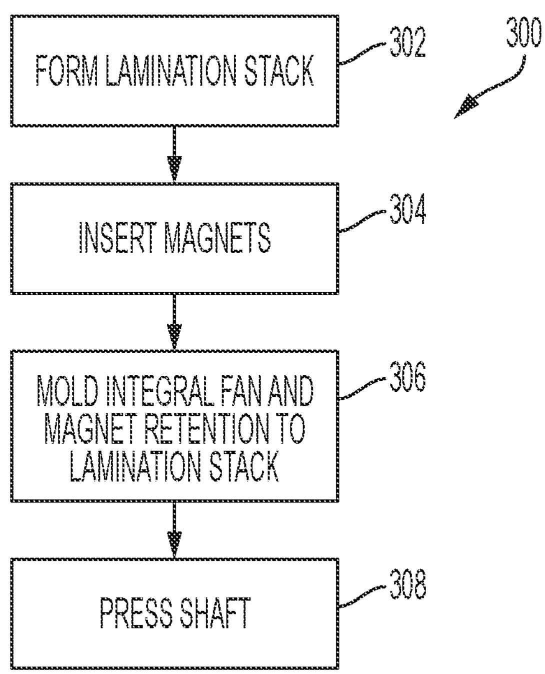

300
FORM LAMINATION STACK
302
INSERT MAGNETS
304
MOLD INTEGRAL FAN AND MAGNET RETENTION TO LAMINATION STACK
306
PRESS SHAFT
308

FIG. 11

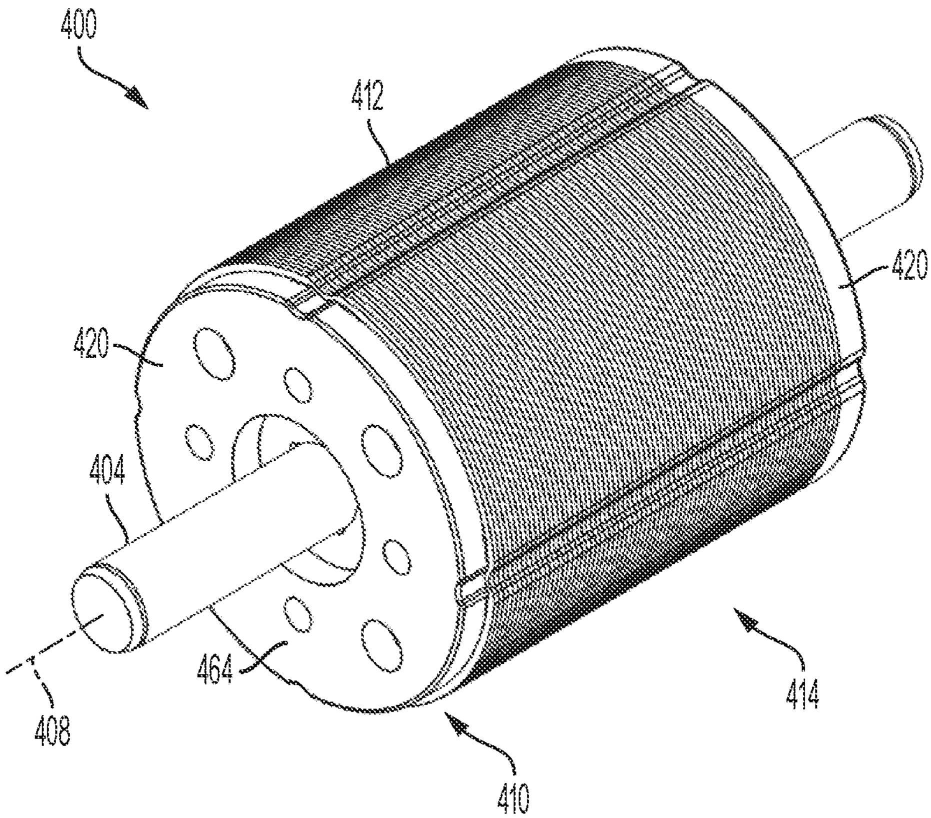


FIG. 12

ROTOR ASSEMBLY FOR AN ELECTRIC MOTOR

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a divisional of U.S. patent application Ser. No. 16/698,283, filed on Nov. 27, 2019, now U.S. Pat. No. 11,742,710, which claims priority to U.S. Provisional Patent Application No. 62/772,924, filed on Nov. 29, 2018, the entire contents of both of which are incorporated herein by reference.

FIELD OF THE INVENTION

The present invention relates to power tools, and more particularly to power tools including electric motors having a molded rotor assembly.

BACKGROUND OF THE INVENTION

Tools, such as power tools, can include an electric motor having a rotor assembly to rotate a shaft and generate a torque output. The rotor assembly may include a fan molded to a lamination stack to form a rotor body, and the shaft may be pressed into the rotor body to form the rotor assembly.

SUMMARY OF THE INVENTION

The present invention provides, in one aspect, a method for forming a molded rotor assembly for an electric motor. The method includes providing a lamination stack, molding an insulative material to the lamination stack to form an integral fan and magnet retention coupled to the lamination stack, the lamination stack and the integral fan and magnet retention together forming a rotor body. The method further includes pressing a shaft into a central aperture formed in the rotor body to achieve a press-fit engagement between the shaft and the lamination stack.

The present invention provides, in another aspect, a method of manufacturing an electric motor. The method includes forming a molded rotor assembly and providing a stator assembly operable to produce a rotating magnetic field with which the rotor assembly interacts. The molded rotor assembly is formed by providing a lamination stack, molding an insulative material to the lamination stack to form an integral fan and magnet retention coupled to the lamination stack, the lamination stack and the integral fan and magnet retention together form a rotor body, and pressing a shaft into a central aperture formed in the rotor body to achieve an interference fit between the shaft and the lamination stack.

Other aspects of the application will become apparent by consideration of the detailed description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is an exploded view of a prior art rotor assembly for an electric motor.

FIG. **2** is a perspective view of a rotor assembly according to an embodiment of the present invention.

FIG. **3** is a perspective view of a rotor body of the rotor assembly of FIG. **2**.

FIG. **4** is another perspective view of the rotor body of FIG. **3**.

FIG. **5** is an end view of a lamination stack of the rotor assembly of FIG. **2**.

FIG. **6** is a perspective view of an integral fan and magnet retention of the rotor assembly of FIG. **2**.

FIG. **7** is and end view of the rotor body of FIG. **3**.

FIG. **8** is a partially exploded view of the rotor assembly of FIG. **2**.

FIG. **9** is another perspective view of the rotor body of FIG. **3**.

FIG. **10** is a detail view of a portion of a rotor body according to another embodiment of the present invention.

FIG. **11** is a flowchart depicting a method of manufacturing a molded rotor assembly for an electric motor.

FIG. **12** is a perspective view of a rotor assembly according to another embodiment of the invention.

Before any embodiments of the application are explained in detail, it is to be understood that the application is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The application is capable of other embodiments and of being practiced or of being carried out in various ways.

DETAILED DESCRIPTION

FIG. **1** illustrates an exploded view of a prior art rotor assembly **10** for an electric motor (not shown). The rotor assembly **10** is supported for rotation with respect to a stator (not shown) and includes a solid shaft **14** that extends along a longitudinal or rotational axis **18**. The rotor assembly **10** also includes a lamination stack **22**, a fan **26**, a rubber ring **30**, and a balance bushing **34**. The lamination stack **22** is formed from a plurality of laminations that are stacked along the rotational axis **18**. The shaft **14** is received into a central aperture (not shown) formed in the lamination stack **22**. The fan **26** is coupled to the shaft **14** adjacent the lamination stack **22** so that the fan **26** rotates with the shaft **14** and provides cooling air to the electric motor. The rubber ring **30** is disposed between the fan **26** and the lamination stack **22**. The balance bushing **34** is coupled to the shaft **14** adjacent the lamination stack **22** and opposite the fan **26** to rotationally balance the rotor assembly **10**.

An outer surface of the shaft **14** includes knurls or splines **38** that engage the central aperture of the lamination stack **22** to rotatably fix the lamination stack **22** to the shaft **14**. Moreover, the central aperture of the lamination stack **22** includes notches (not shown) that are used for orientation of parts for magnetization of magnets during the assembly process. In the prior art rotor assembly **10**, imperfect knurls formed on the shaft **14** combined with the notches in the lamination stack **22** can be a source of imbalance in the rotor assembly **10**. Thus, the balance bushing **34** is required to balance the rotor assembly **10**.

FIGS. **2-9** illustrate a molded rotor assembly **100** (and portions thereof) for an electric motor (not shown) according to the present invention. The electric motor may be used in various different tools, such as power tools (e.g., rotary hammers, pipe threaders, cutting tools, etc.), outdoor tools (e.g., trimmers, pole saws, blowers, etc.), and other electrical devices (e.g., motorized devices, etc.).

The electric motor is configured as a brushless DC motor. In some embodiments, the motor may receive power from an on-board power source (e.g., a battery, not shown). The battery may include any of a number of different nominal voltages (e.g., 12V, 18V, etc.), and may be configured having any of a number of different chemistries (e.g., lithium-ion, nickel-cadmium, etc.). Alternatively, the motor may be powered by a remote power source (e.g., a household electrical outlet) through a power cord. The motor includes a substantially cylindrical stator (not shown) operable to produce a magnetic field. The rotor assembly **100** is rotatably supported by a solid shaft **104** and configured to co-rotate with the shaft **104** about a longitudinal or rotational axis **108**.

The rotor assembly **100** includes an integral fan and magnet retention **110** or main body (FIG. **6**) formed of an insulative material (e.g., plastic) that is molded to a lamination stack **112** to form a rotor body **114**. The integral fan and magnet retention **110** includes a fan portion **116** and a magnet retention portion **120** formed opposite the fan portion **116**. When the integral fan and magnet retention **110** is molded to the lamination stack **112**, the fan portion **116** abuts one end of the lamination stack **112** to define a fan end **124** of the rotor body **114**, and the magnet retention portion **120** abuts an opposite end of the lamination stack **112** to define a magnet retention end **128**.

The lamination stack **112** defines a longitudinally extending central aperture **132** that receives the shaft **104** by press-fit engagement. Magnet slots **136** (FIG. **5**) are formed in the lamination stack **112** and configured to receive permanent magnets (not shown). The lamination stack **112** also includes injection channels **140** formed about the central aperture **132** and extending longitudinally between the fan end **124** and magnet retention end **128**. When the integral fan and magnet retention **110** is molded to the lamination stack **112**, the insulative material of the integral fan and magnet retention **110** flows through the channels **140** and joins the fan portion **116** to the magnet retention portion **120**. The insulative material also extends around the magnets within the magnet slots **136** to form magnet holding portions **144** (FIG. **6**). The magnet holding portions **144** extend through the magnet slots **136** between the fan portion **116** and the magnet retention portion **120**, and surround the permanent magnets to retain the magnets within the slots **136**.

The rotor body **114** is secured to the shaft **104** by interference fit (e.g., by press-fit) to form the molded rotor assembly **100**. With reference to FIGS. **7** and **8**, unlike the prior art shaft **14** having splines **38** described above, the shaft **104** of the present invention includes a smooth annular outer surface **148**. In the illustrated embodiment, the smooth annular outer surface **148** is cylindrical and devoid of splines or other retention features. The central aperture **132** of the lamination stack **112** is partially defined by press-fit portions **152** (FIG. **7**) that contact and engage the smooth annular outer surface **148** of the shaft **104** to transfer torque between the rotor body **114** and the shaft **104**. The central aperture **132** is further defined by relief notches **156** formed in the lamination stack **112** between adjacent press-fit portions **152** to relieve stresses during the pressing process. By providing the shaft **104** with the smooth annular outer surface **148** and pressing the shaft **104** into the central aperture **132**, the rotor assembly **100** of the present invention eliminates the imbalance issue associated with the prior art splines **38**. Thus, the rubber ring **30** and the balance bushing **34** of the prior art rotor assembly **10** are eliminated in the molded rotor assembly **100**.

In known prior art electric motors in which the fan is molded to the lamination stack and the shaft engages the lamination stack by press-fit, the shaft is pressed into the lamination stack prior to the molding process. In the molded rotor assembly **100** of the present invention, the shaft **104** is pressed into the rotor body **114** after the integral fan and magnet retention **110** is molded to the lamination stack **112**. This avoids the costs of having many sets of molding inserts for different shaft sizes and reduces the cost of the shaft **104** itself.

The shaft **104** is pressed into the rotor body **114** from the fan end **124** as indicated by the arrow shown in FIG. **8**, and the rotor body **114** is supported at the magnet retention end **128** during pressing. The fan portion **116** and the magnet retention portion **120** each include shaft openings **160** (FIG. **6**) that correspond to the central aperture **132** to permit the shaft **104** to pass therethrough. With reference to FIG. **9**, the magnet retention portion **120** defines a bearing surface **164**, and the rotor body **114** is supported at the bearing surface **164** as the shaft **104** is pressed into the rotor body **114** from the fan end **124**. In other embodiments (not shown), the bearing surface may alternatively be provided on the fan portion **116**. In such embodiments, the shaft **104** may be pressed into the rotor body **114** from the magnet retention end **128** (i.e., in a direction opposite to the arrow shown in FIG. **8**). A fixture (not shown) may be employed to support the rotor body **114** during pressing.

FIG. **10** illustrates another embodiment of a molded rotor assembly **200** similar to the molded rotor assembly **100** described above, with like features shown with reference numerals plus "100." The rotor assembly **200** also includes a lamination stack **212** and an integral fan and magnet retention **210**, and a shaft **204** that is pressed into a central aperture **232** formed in the lamination stack **212**. To avoid a risk of cracks developing in the magnet retention portion **220** during pressing, the magnet retention portion **220** may include an oversized shaft opening **268** (FIG. **10**) that exposes an alternative bearing surface **272** located on the lamination stack **212**. The central aperture **232** may be of a first diameter D**1**, measured between the press-fit portions **252**, while the oversized shaft opening **268** may be of a second diameter D**2** larger than D**1**. The rotor body **214** is supported at the alternative bearing surface **272** of the lamination stack **212** during pressing while the shaft **204** is pressed from the fan end **224**. In other embodiments (not shown), the oversized shaft opening may alternatively be provided in the fan portion, so that the alternative bearing surface is located at the fan end **224**. In such embodiments, the shaft **204** may be pressed from the magnet retention end while the rotor body **214** is supported at the alternative bearing surface at the fan end **224**.

FIG. **11** illustrates a method **300** of manufacturing a rotor assembly for an electric motor according to the present invention. In general, the illustrated method **300** includes a step **302** to form a lamination stack, a step **304** to insert permanent magnets into magnet slots formed in the lamination stack, a step **306** to mold an integral fan and magnet retention to the lamination stack to form a rotor body, and a step **308** to press a shaft into a central aperture formed in the rotor body. The method of FIG. **11** differs from prior art methods in that the pressing occurs at step **308** after the integral fan and magnet retention is molded to the lamination stack at step **306**. In some embodiments, the process may omit one or more of the steps **302** and **304** yet still fall within the scope of the present invention.

FIG. **12** illustrates another embodiment of a molded rotor assembly **400** similar to the molded rotor assemblies **100** described above, with like features shown with reference numerals plus "300." The rotor assembly **400** includes a shaft **404** rotatable about a longitudinal or rotational axis **408**, and a rotor body **414** secured to the shaft **404** (e.g., by interference fit via the pressing method described above). The rotor body **414** includes a lamination stack **412** and an integral magnet retention **410** or main body (FIG. **12**). The integral magnet retention **410** is formed of an insulative material (e.g., plastic) that is molded to the lamination stack **412** to form the rotor body **414**.

Unlike the integral fan and magnet retentions **110**, **210** described above, the integral magnet retention **410** does not include a fan portion. Instead, the integral magnet retention **410** includes a pair of magnet retention portions **420** abutting each axial end of the lamination stack **412**. In the illustrated embodiment, the two magnet retention portions **420** are identical. When the integral magnet retention **410** is molded to the lamination stack **412**, the insulative material of the magnet retention **410** flows through channels (not shown) formed in the lamination stack **412** and joins the two respective magnet retention portions **420**. The insulative material also extends around the magnets within the magnet slots (not shown) to form magnet holding portions (not shown), similar to that described above with respect to FIG. **6**. The magnet holding portions extend through the magnet slots between the two magnet retention portions **420**, and surround the permanent magnets to retain the magnets within the slots. Each magnet retention portion **420** also defines a bearing surface **464**, and the rotor body **414** can be supported at the bearing surface **464** of either of the two magnet retention portions **420** as the shaft **404** is pressed into the rotor body **414**.

Although the application has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the scope and spirit of one or more independent aspects of the application as described.

What is claimed is:

1. A method of forming a molded rotor assembly for an electric motor, the method comprising:
- providing a lamination stack;
- molding an insulative material to the lamination stack to form an integral fan and magnet retention coupled to the lamination stack, the lamination stack and the integral fan and magnet retention together forming a rotor body; and
- pressing a shaft into a central aperture formed in the rotor body to achieve an interference fit between the shaft and the lamination stack.

2. The method of claim **1**, wherein the shaft is pressed into the rotor body after the insulative material is molded to the lamination stack.

3. The method of claim **1**, wherein the lamination stack defines a plurality of magnet slots configured to receive permanent magnets, and wherein the method further comprises inserting the permanent magnets into the magnet slots prior to molding the insulative material to the lamination stack.

4. The method of claim **1**, wherein the integral fan and magnet retention includes a fan portion and a magnet retention portion abutting opposite ends of the lamination stack, the magnet retention portion defining a bearing surface, and wherein the method further includes supporting the rotor body at the bearing surface prior to pressing the shaft into the central aperture.

5. The method of claim **1**, wherein the lamination stack defines the central aperture and the shaft includes an outer surface, and wherein pressing the shaft into the central aperture includes engaging the outer surface of the shaft and the central aperture of the lamination stack.

6. The method of claim **5**, wherein the outer surface of the shaft is cylindrical.

7. The method of claim **5**, wherein the central aperture has a size smaller than the outer surface of the shaft.

8. The method of claim **7**, wherein the integral fan and magnet retention includes an oversized shaft opening larger than the central aperture and the outer surface of the shaft.

9. The method of claim **5**, wherein the central aperture has an inner diameter smaller than an outer diameter of the shaft.

10. The method of claim **1**, wherein the insulative material is plastic.

11. The method of claim **1**, wherein the central aperture of the lamination stack includes a plurality of press-fit portions and a plurality of relief notches formed between adjacent press-fit portions, the relief notches being configured to relieve stress while the shaft is pressed into the central aperture.

12. A method of manufacturing an electric motor, the method comprising:
- forming a molded rotor assembly by
  - providing a lamination stack,
  - molding an insulative material to the lamination stack to form an integral fan and magnet retention coupled to the lamination stack, the lamination stack and the integral fan and magnet retention together forming a rotor body, and
  - pressing a shaft into a central aperture formed in the rotor body to achieve an interference fit between the shaft and the lamination stack; and
- providing a stator assembly operable to produce a rotating magnetic field with which the rotor assembly interacts.

13. The method of claim **12**, wherein the shaft is pressed into the rotor body after the insulative material is molded to the lamination stack.

14. The method of claim **12**, further comprising locating the rotor assembly relative to the stator assembly such that the rotating magnetic field produced by the stator assembly imparts torque to the rotor assembly.

15. The method of claim **12**, further comprising inserting the rotor assembly within the stator assembly to create an inner rotor electric motor.

16. The method of claim **12**, wherein the lamination stack defines a plurality of magnet slots configured to receive permanent magnets, and wherein the method further comprises inserting the permanent magnets into the magnet slots prior to molding the insulative material to the lamination stack.

17. The method of claim **12**, wherein during molding of the insulative material to the lamination stack, the insulative material flows through an injection channel of the lamination stack.

18. The method of claim **17**, wherein the injection channel extends longitudinally between a fan end adjacent a fan portion of the integral fan and magnet retention and a magnet retention end adjacent a magnet retention portion of the integral fan and magnet retention.

19. The method of claim **12**, wherein the insulative material is plastic.

20. The method of claim **12**, wherein the central aperture of the lamination stack includes a plurality of press-fit portions and a plurality of relief notches formed between adjacent press-fit portions, the relief notches being configured to relieve stress while the shaft is pressed into the central aperture.

* * * * *